

Final $\mathbf{F}$ and Σ

Inserting

$$(S_t, R_{1,t}, R_{1,0,t}, R_{2,t}, R_{2,0,t})^\top = (X_t^{(1)}, X_t^{(2)}, X_t^{(3)}, X_t^{(4)}, X_t^{(5)})^\top$$

and

$$R_{1,t} = (1 - \theta_1)R_{1,0,t} + \theta_1 R_{1,1,t} \Leftrightarrow R_{1,1,t} = (R_{1,t} - (1 - \theta_1)R_{1,0,t}) \frac{1}{\theta_1}$$

$$R_{2,t} = (1 - \theta_2)R_{2,0,t} + \theta_2 R_{2,1,t} \Leftrightarrow R_{2,1,t} = (R_{2,t} - (1 - \theta_2)R_{2,0,t}) \frac{1}{\theta_2}$$

yields

$$\mathbf{F} = \begin{pmatrix} 0 \\ (1 - \theta_1)X_t^{(3)}(\lambda_{1,1} - \lambda_{1,0}) - \lambda_{1,1}X_t^{(2)} \\ -\lambda_{1,0}X_t^{(3)} \\ \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right)^2 ((1 - \theta_2)\lambda_{2,0} + \theta_2 \lambda_{2,1}) + (1 - \theta_2)X_t^{(5)}(-\lambda_{2,1} - \lambda_{2,0}) - \lambda_{2,1}X_t^{(4)} \\ \lambda_{2,0} \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right)^2 - \lambda_{2,0}X_t^{(5)} \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right) X_t^{(1)} \\ ((1 - \theta_1)\lambda_{1,0} + \theta_1 \lambda_{1,1}) \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right) \\ \lambda_{1,0} \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right) \\ 0 \\ 0 \end{pmatrix}$$

2.2 Separation of parameters

Drift parameters

$$\boldsymbol{\theta} = \{\beta_0, \beta_1, \beta_2, \theta_1, \theta_2, \lambda_{1,0}, \lambda_{1,1}, \lambda_{2,0}, \lambda_{2,1}\}$$

$\mathbf{F}$ function:

$$\mathbf{F} = \begin{pmatrix} 0 \\ (1 - \theta_1)X_t^{(3)}(\lambda_{1,1} - \lambda_{1,0}) - \lambda_{1,1}X_t^{(2)} \\ -\lambda_{1,0}X_t^{(3)} \\ \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right)^2 ((1 - \theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1}) + (1 - \theta_2)X_t^{(5)}(-\lambda_{2,1} - \lambda_{2,0}) - \lambda_{2,1}X_t^{(4)} \\ \lambda_{2,0} \left(\beta_0 + \beta_1 X_t^{(2)} + \beta_2 \sqrt{X_t^{(4)}}\right)^2 - \lambda_{2,0}X_t^{(5)} \end{pmatrix}$$

Volatility parameters

$$\boldsymbol{\theta} = \{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\beta}_2, \tilde{\theta}_1, \tilde{\theta}_2, \tilde{\lambda}_{1,0}, \tilde{\lambda}_{1,1}, \tilde{\lambda}_{2,0}, \tilde{\lambda}_{2,1}\}$$

Σ function:

$$\Sigma = \begin{pmatrix} \left(\tilde{\beta}_0 + \tilde{\beta}_1 X_t^{(2)} + \tilde{\beta}_2 \sqrt{X_t^{(4)}}\right) X_t^{(1)} \\ ((1 - \tilde{\theta}_1)\tilde{\lambda}_{1,0} + \tilde{\theta}_1\tilde{\lambda}_{1,1}) \left(\tilde{\beta}_0 + \tilde{\beta}_1 X_t^{(2)} + \tilde{\beta}_2 \sqrt{X_t^{(4)}}\right) \\ \tilde{\lambda}_{1,0} \left(\tilde{\beta}_0 + \tilde{\beta}_1 X_t^{(2)} + \tilde{\beta}_2 \sqrt{X_t^{(4)}}\right) \\ 0 \\ 0 \end{pmatrix}$$

3 Three dimensional Brownian Motions

$$W_t = (W_t^{(1)}, W_t^{(2)}, W_t^{(3)})$$

$$\frac{dS_t}{S_t} = \sigma_t dW_t^{(1)}$$

$$\sigma_t = \sigma(R_{1,t}, R_{2,t}) = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}$$

$$R_{1,t} = (1 - \theta_1) R_{1,0,t} + \theta_1 R_{1,1,t}$$

$$R_{2,t} = (1 - \theta_2) R_{2,0,t} + \theta_2 R_{2,1,t}$$

The sub-factors evolve according to:

$$\begin{aligned} dR_{1,0,t} &= \lambda_{1,0} \left(\frac{dS_t}{S_t} - R_{1,0,t} dt \right) = -\lambda_{1,0} R_{1,0,t} dt + \lambda_{1,0} \sigma(R_{1,t}, R_{2,t}) dW_t^{(2)} \\ dR_{1,1,t} &= \lambda_{1,1} \left(\frac{dS_t}{S_t} - R_{1,1,t} dt \right) = -\lambda_{1,1} R_{1,1,t} dt + \lambda_{1,1} \sigma(R_{1,t}, R_{2,t}) dW_t^{(3)} \\ dR_{2,0,t} &= \lambda_{2,0} \left(\left(\frac{dS_t}{S_t} \right)^2 - R_{2,0,t} dt \right) = \lambda_{2,0} (\sigma^2(R_{1,t}, R_{2,t}) - R_{2,0,t}) dt \\ dR_{2,1,t} &= \lambda_{2,1} \left(\left(\frac{dS_t}{S_t} \right)^2 - R_{2,1,t} dt \right) = \lambda_{2,1} (\sigma^2(R_{1,t}, R_{2,t}) - R_{2,1,t}) dt \end{aligned}$$

We now need to find our $\Sigma(\mathbf{X}_t, \boldsymbol{\theta})$. We observe that it will be a 5×5 matrix.

$$\begin{aligned} & \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}_{5 \times 5} \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}_{5 \times 1} = \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}_{5 \times 1} \\ & \begin{pmatrix} \sigma_t S_t & 0 & 0 & 0 & 0 \\ 0 & ((1 - \theta_1) \lambda_{1,0} + \theta_1 \lambda_{1,1}) \sigma_t & 0 & 0 & 0 \\ 0 & 0 & \lambda_{1,0} \sigma_t & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} dW_t^{(1)} & dW_t^{(2)} & dW_t^{(3)} & 0 & 0 \end{pmatrix} \\ & = \begin{pmatrix} \sigma_t S_t dW_t^{(1)} \\ ((1 - \theta_1) \lambda_{1,0} + \theta_1 \lambda_{1,1}) \sigma_t dW_t^{(2)} \\ \lambda_{1,0} \sigma_t dW_t^{(3)} \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$